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# Supplementary Materials for

## Gate-tunable van der Waals heterostructure for reconfigurable neural network vision sensor

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Figs. S1 to S10

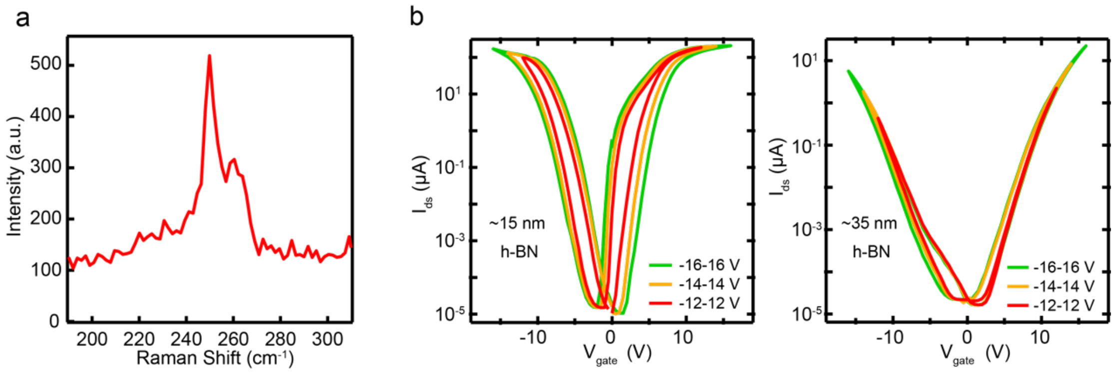


**Fig. S1 Raman and electrical characterization of $WSe_2$ devices. (a)** Raman spectrum of $WSe_2$. The excitation wavelength of the laser is 514 nm. (b) The transfer curves of $WSe_2$ devices for h-BN with different thickness (15 and 35 nm thick). The hysteresis window in the transfer curves of the thin h-BN may derive from the charge trapping in the h-BN during sweeping back gate voltages. Note that the field effect characteristics of the thick $WSe_2$ devices are ambipolar and are different from the p-type transport characteristic of thinner $WSe_2$ devices. This difference has been reported before and attributed to the thickness difference of $WSe_2$.

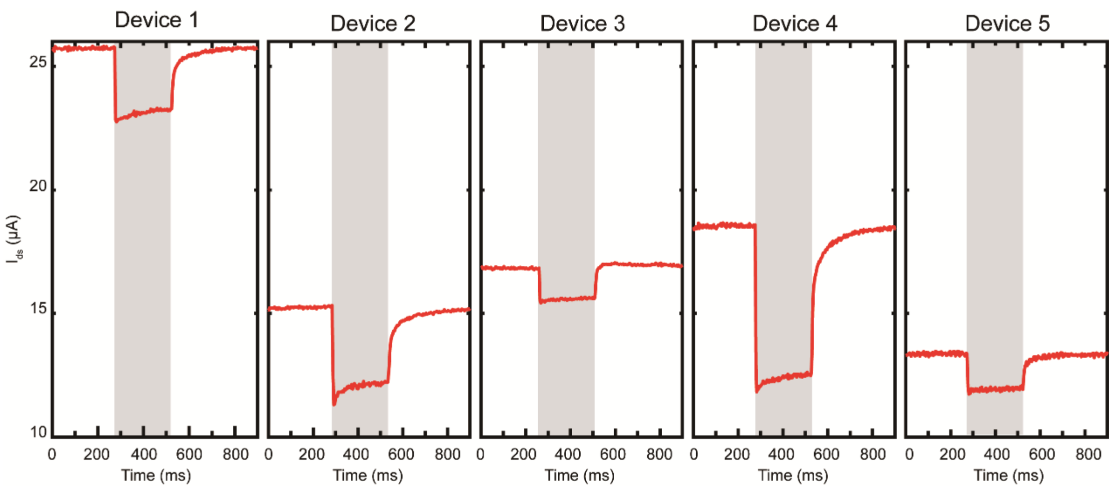


**Fig. S2 Device variance of OFF-photoresponse in 5 different devices under light illumination at $V_g$= -12 V and $V_{ds}$=0.1 V**. The light intensity is 12.6 nW/μm$^2$. Performance of these devices differ in dark current, response time and current reduction. Shadow areas represent the duration of light illumination. The variances may come from fabrication process, such as variation in the thickness of $WSe_2$ and h-BN by mechanical exfoliation. Optimizing fabrication process might alleviate this issue.

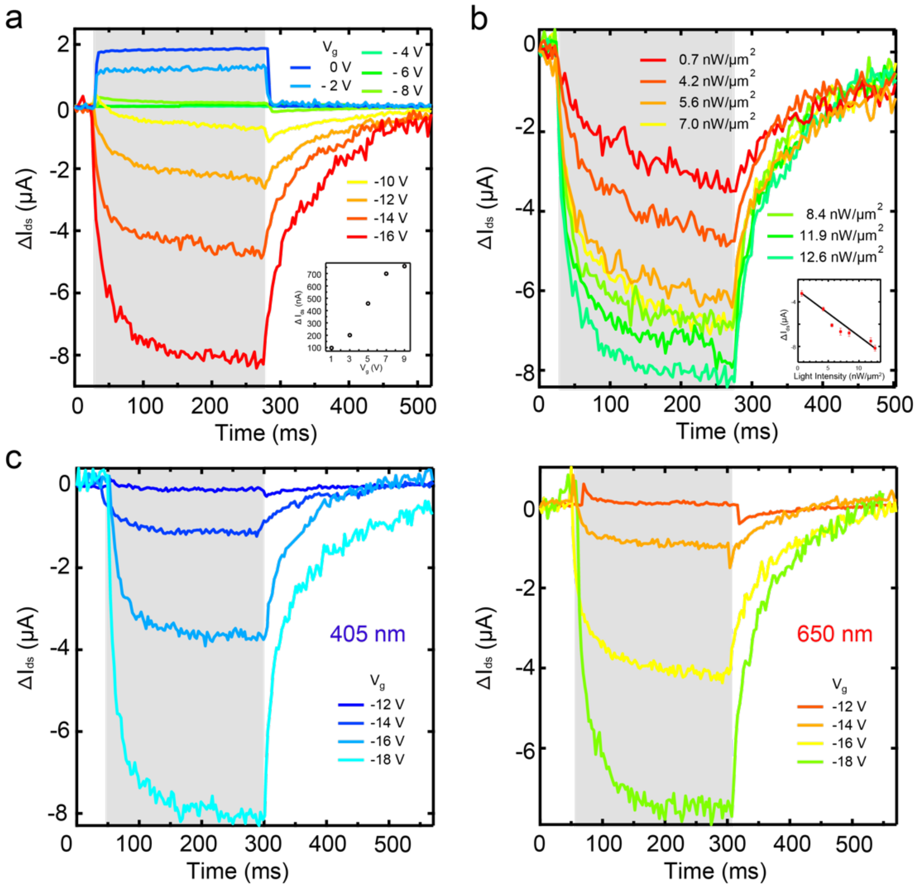


**Fig. S3 Photoresponse of the $WSe_2$/BN heterostructure device under different gate voltages, light intensities and wavelengths.** (a) Photoresponse at different gate voltages. The inset shows the current variation as a function of positive gate voltage. A wavelength of 405 nm with a light intensity of 12.6 nW/μm$^2$ was used in the experiment. With varying negative gate voltages, we can achieve a transition between the positive and negative photoresponses of the vdW heterostructure device, which has been used in the main text to emulate the functionality of the artificial receptive field (RF) of the human retina. (b) Photoresponse of the vdW heterostructure device is enhanced by increasing intensity of 405 nm light at $V_g$= -16 V. Insert is the linear relation between photoresponse and light intensity. (c) Photoresponse of the vdW heterostructure device at different wavelengths. Both purple (405 nm) and red (650 nm) lights give rise to a negative photoresponse, indicating that the vdW heterostructure device could operate within the whole visible light range. Shadow areas represent the duration of light illumination.

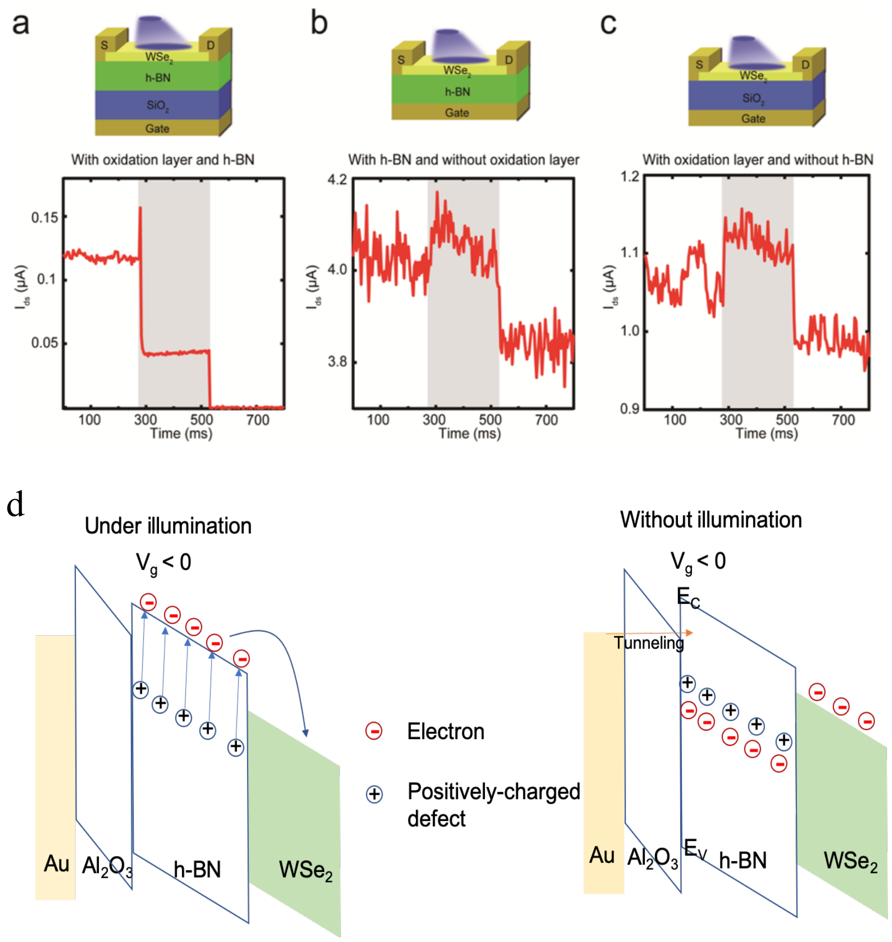


**Fig. S4 Photoresponse of the devices with and without h-BN/oxidation layer dielectric material as well as the proposed working principle of the vertical heterostructure device.** (a) With both h-BN (~40 nm) and oxidation layer ($SiO_2$ ~280 nm), light illumination leads to a reduction in current and the reduced current is retained at a constant value. (b) With only h-BN (~40 nm), the light illumination increases the conductivity and leads to positive photoresponse. (c) With only $SiO_2$ (~280 nm), positive photoresponse can be observed similar to (b). Light with 405 nm and 12.6 nW/μm$^2$ was used in the experiment, and shadow areas represent the duration of light illumination. (d) Possible mechanism of the OFF device with and without the light illumination. To elucidate the role of vacancy defects in the Off photoresponse of the vertical heterostructure device, we have used one defect energy level to represent all the possible defects close to conduction band of h-BN.

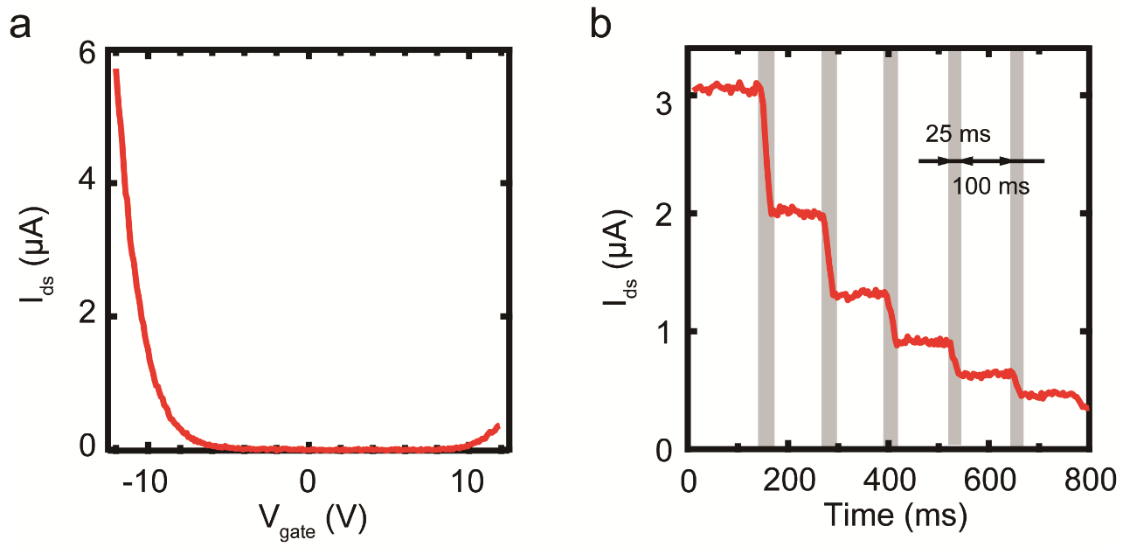


**Fig. S5 Effect of light illumination on the electrical characteristics of $WSe_2$ device.** (a) Transfer curve of $WSe_2$ without light illumination (b) Current variance of $WSe_2$ device under light illumination at $V_g$ = -12 V. The thickness of h-BN and $Al_2O_3$ is ~40 nm and ~10 nm, separately. In the experiment, light pulses with 25 ms duration and 100 intervals (405 nm and 140 pW/μm$^2$) were applied to the device. The light illumination leads to a shift in the transfer curve. Threshold voltage of the shift under each light pulse can be obtained by extracting the values of platforms in $I_{ds}$ vs Time curve and corresponding values of $V_g$ in transfer curve. The photodoping rate is expressed as: $R_{doping\ rate} = \frac{\Delta n}{\Delta t}$. Where, Δt is hold time of light pulse, ~25 ms; $\Delta n = C \times \Delta V_{g\text{-}shift}/e$ and $\Delta V_{g\text{-}shift}$ is measured to be 0.5 V; C is the capacitance of parallel plate capacitor based on h-BN/$Al_2O_3$ heterostructure (~$10^{-14}$ F). Thus, $R_{doping\ rate}$ is estimated around $10^{13}$ cm$^{-2}$s$^{-1}$.

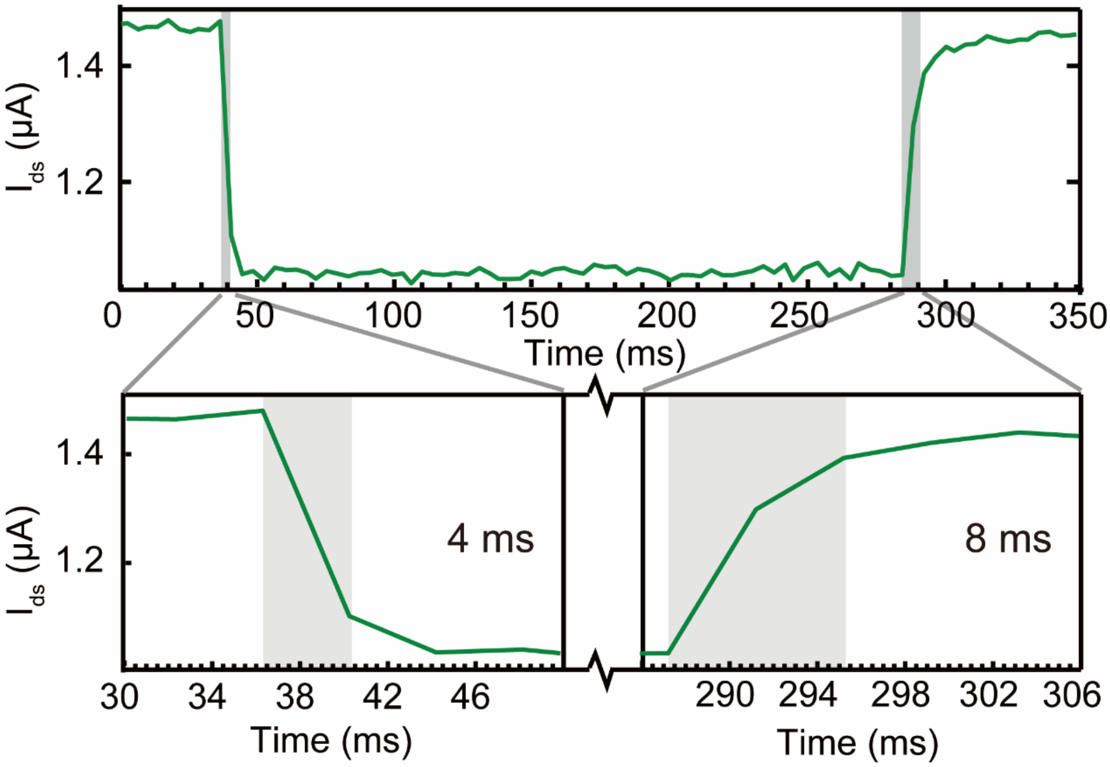


**Fig. S6 Response time of OFF-photoresponse device based on thin materials under $V_g$ = -9 V.** The response time can be reduced down to less than 8 ms by optimizing device parameters ($WSe_2$ ~2 nm, h-BN ~10 nm, $Al_2O_3$ ~6 nm) and fabrication process. Shadow areas denote the rising and falling time. Light intensity is 4.2 nW/μm$^2$. Note that the falling and rising time of the thin materials device are considerably reduced compared to that based on the thick materials ($WSe_2$ ~20 nm, h-BN ~40 nm, $Al_2O_3$ ~10 nm), which helps increasing the operating speed of the device. We also note that the dark current of retinomorphic devices can be affected by the thickness of the $WSe_2$. Thinning the $WSe_2$ down to single layer reduces the dark current and the power consumption of the retinomorphic devices due to the increased Schottky barrier.

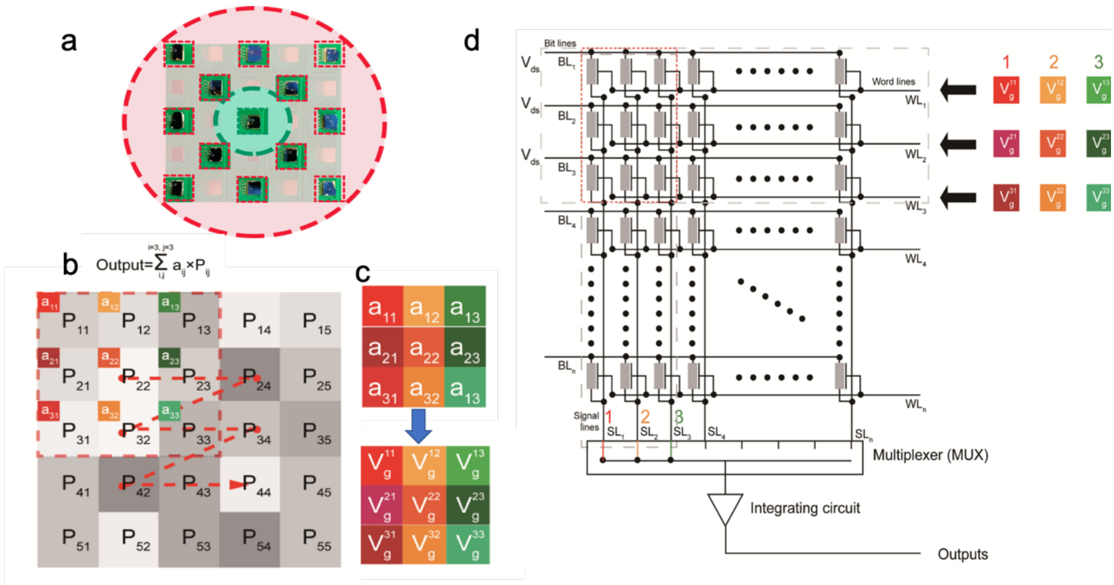


**Fig. S7 Optical image of the vision sensor and circuit design for large-scale retinomorphic array.** (a) Each pixel in the vision sensor is placed in a socket mounted on a PCB board and connected in parallel. The pixel in green area is OFF-photoresponse device and the rest of them are ON-photoresponse devices. (b) Illustration of convolution for image. $P_{ij}$ represents pixel values in the image (5×5) and $a_{ij}$ represents kernel values (3×3). The output is a summation of $\sum_{i,j}^{i=3,j=3} a_{ij} \times P_{ij}$. The kernel would move along the dotted line with arrow to carry out the convolution computation of the whole image. (c) Converting kernel values into gate voltages values ($V_g^{ij}$, 3×3) according to the photoresponse of the retinomorphic devices. (d) Circuit scheme for large-scale retinomorphic array. Gray rectangle represents the retinomorphic device. To achieve large-scale circuit design of retinomorphic array, we could separate kernel into several column vectors and make a summation after feeding the corresponding column vector of $V_g$ into the retinomorphic array one by one.

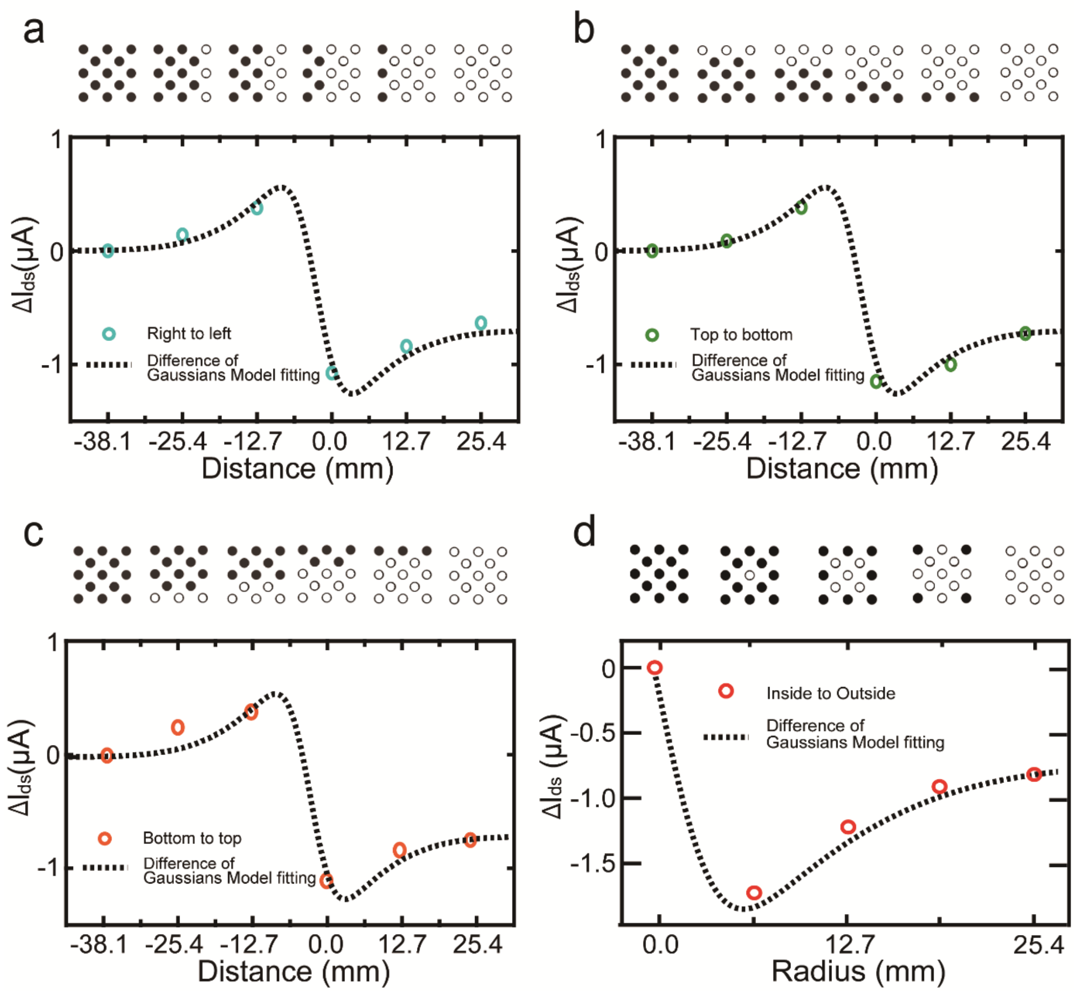


**Fig. S8 Response of the vision sensor as an artificial receptive field (RF) for detecting a contrast-reversing edge moving along different directions**. (a) From right to left. (b) From top to bottom. (c) From bottom to top. The results indicate that the photoresponse of the vision sensor is isotropic in different directions. (d) From inside to outside, the photocurrent decreases drastically as the radius of the light spot increases until a dip. With a further increase in the radius, the photoresponse becomes saturated.

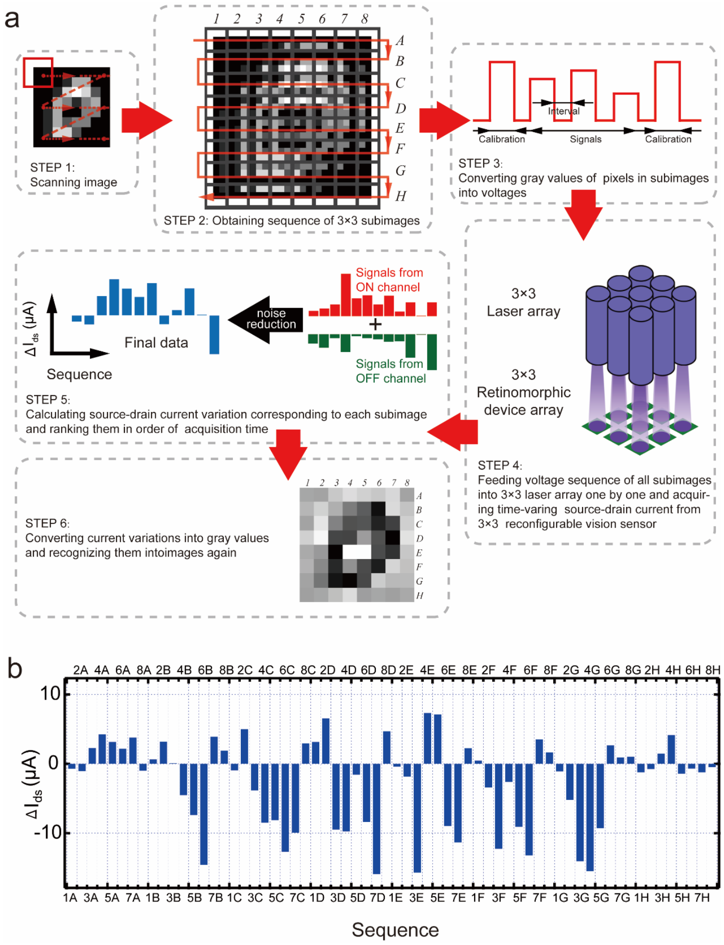


**Fig. S9 Two-dimensional filter implemented with the vision sensor**. (a) The setup for carrying out two-dimensional filtering or convolution. We used the vision sensor to scan every image and split them into many subimages (3×3) with the sequence as shown in red arrow (STEP 1). Then, we converted pixel values in each subimages to voltages sequence. 64 subimages are in 8×8 array, 1-8 and A-H representing column and row coordinates (STEP 2). After that, voltage signals sequence (STEP 3) were fed into laser array (3×3) to generate light patterns sequence (STEP 4). At the same time, we measured current signals from the ON and OFF channels following with noise reduction, which correspond to the source/drain current of ON-/OFF-photoresponse devices (STEP 5). Finally, these measured data on current variation can be ranked and reorganized into images as recognized results (STEP 6). The pixels in “0” in the STEP 6 correspond to the convolution results in subimages of STEP 2 with the same coordinates. (b) The measured current variation of numeral “0” as a function of scanning sequence along the red arrow indicated in the STEP 1.

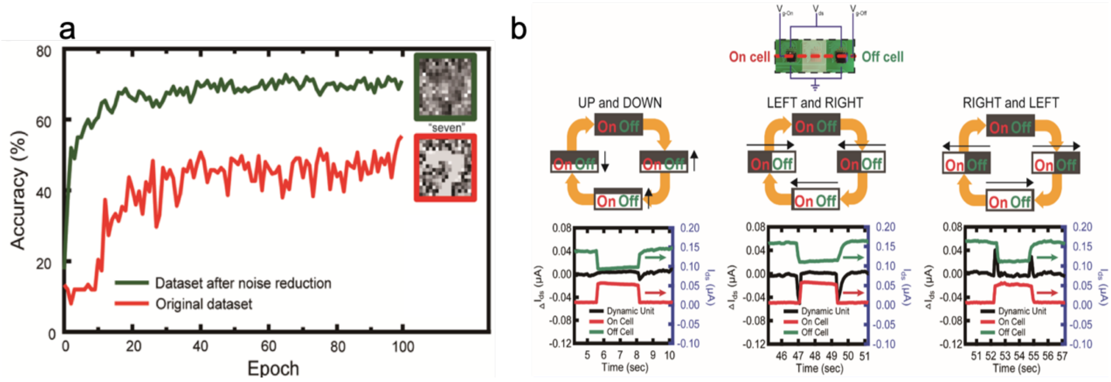


**Fig. S10 The operation of noise reduction (a) and real-time tracking edge (b) by our vision sensor.** Recognition accuracy of the original dataset is compared to that of the dataset after noise reduction in the same neural network setting. Inserts represent the original image "7" and image filtered by the retinomorphic device array and the kernel used for noise reduction. The Keras and TensorFlow were used for artificial neural network training. Ten percent of the datasets was chosen as test sets. Prototype demonstration of the real-time motion detecting based on a dynamic unit connecting two retinomorphic devices in parallel. The top panel shows the dynamic unit made of an On device and an Off device. The middle panel shows the flow charts of detecting motion along different directions. The bottom panel shows the measured current variations of each retinomorphic device (On and Off) and dynamic unit in response to the different motions.